Vishal Kaushal

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Professional Summary

Applied **AI Engineer** building production **Generative AI** and **Speech AI** systems: an **AI dubbing platform** on AWS, an emotion-aware **Speech-to-Speech Translation** pipeline, **RAG**, and **LoRA fine-tuning**, using **Python, FastAPI, and Docker**.

Technical Skills

- **Gen AI & Speech:** RAG, LangChain, ChromaDB, FAISS, LoRA Fine-tuning, HuggingFace Transformers, WhisperX, TTS & Voice Cloning (ElevenLabs, MOSS-TTS, RVC)
- **Machine Learning:** PyTorch, TensorFlow, Scikit-Learn, OpenCV, MediaPipe, YOLO, librosa
- **Backend:** Python, FastAPI, REST APIs, Server-Sent Events, SQLAlchemy, Alembic, PostgreSQL, SQLite
- **Cloud & DevOps:** AWS (App Runner, S3, ECR, CloudWatch), Docker, Vercel, Git, ffmpeg
- **Frontend:** TypeScript, React, Vite, Tailwind CSS

Professional Experience

R&D Engineer **Feb 2026 – Present**
Rikaian Technology Pvt. Ltd. *Pune*

- Built an **end-to-end AI dubbing platform** on the ElevenLabs Dubbing API (FastAPI + React, AWS App Runner, S3, PostgreSQL): upload, auto-transcription, per-line translation editing, re-dubbing in **100+ languages**, and export.
- Cut dubbing cost on long, silence-heavy tracks by **sending only detected speech** to the API and re-inserting dubbed lines at their original timestamps, so output length and timing match the source.
- Fixed recurring **out-of-memory crashes** in production by streaming audio processing, lowering peak memory from **1.9 GB to 0.49 GB** on a 21-minute track.
- Reduced stored audio size by **95%** by storing lossless FLAC instead of WAV, with WAV/MP3 conversion on download.
- Architected an emotion-aware **Speech-to-Speech Translation (S2ST)** pipeline (English ↔ Chinese/Hindi) with WhisperX, emotion2vec+, and MOSS-TTS 8B, tagging sentence-level emotion and word-level emphasis.
- Engineered a **LoRA fine-tuning workflow** for MOSS-TTS 8B on a Colab A100 with a 322-clip dataset, enabling emotion-controlled voice cloning.
- Designed a **DTW-based emotion preservation score** (F0 contour, energy, rhythm) to compare TTS outputs across ElevenLabs and MOSS-TTS.

AI Intern **Jul 2024 – Sep 2024**
EduSkill Foundation *Remote*

- Engineered a mobile-optimized **object detection** module running at **30 FPS** for real-time visual analysis.
- Integrated Google Cloud Vision API into an Android app for visual search with **92% recognition accuracy**.
- Applied model **quantization and pruning** to reduce latency on resource-limited mobile hardware.

Key Projects

Legal RAG API | *FastAPI, LangChain, ChromaDB, Docker* GitHub

- Deployed a containerized **RAG API** answering Indian Constitutional, Criminal, and Labour law questions, with **domain routing** that sends each query to the right vector store.
- Returned the **cited legal sections** with every answer to ground responses and limit hallucination.
- Fit free-tier RAM/timeout limits with a custom Docker image (CPU-only PyTorch, pre-baked model weights).

Full-Duplex Voice AI Agent | *Python, LiveKit, WebRTC* GitHub

- Built a low-latency **voice-to-voice agent** (STT → LLM → TTS over WebRTC) with **sub-500 ms** end-to-end latency.
- Added **voice activity detection** so users can interrupt ("barge in") and the agent stops speaking instantly.

Publications

- V. Kaushal, P. Raut, S. Amalraj, "Review of Home Automation Using ML for Gesture Control and Safety Detection," *IJDSAI*, 2024. [Link]

Education

B.E. in Computer Engineering **2021 – 2025**
KJ College of Engineering and Management Research (KJCOEMR), Pune *CGPA: 8.04 / 10*